

UNIVERSITY
OF MIAMI



Variation-Aware Evaluation of Haitian Creole Medical Translations: A Step Toward Inclusive NLP for Healthcare

LASSO 2025

Ludovic Mompelat, Assistant Professor

Rose Flaure Accilien, PhD Student

University of Miami, Modern Languages and Literatures

Haitian Creole in Miami: Context

UNIVERSITY
OF MIAMI



1. Haitian Creole = 3rd most spoken language in Florida
 2. Miami has one of the largest diaspora communities outside Haiti
 3. Many speakers are Limited English Proficient (LEP)
 4. Healthcare settings often lack adequate interpretation resources
- Proposal : linguistically-informed NLP approach to a culturally-accurate AI-assisted medical translation system (Mompelat 2025)
 - In collaboration with Dr. David Tézil (University of Alabama)

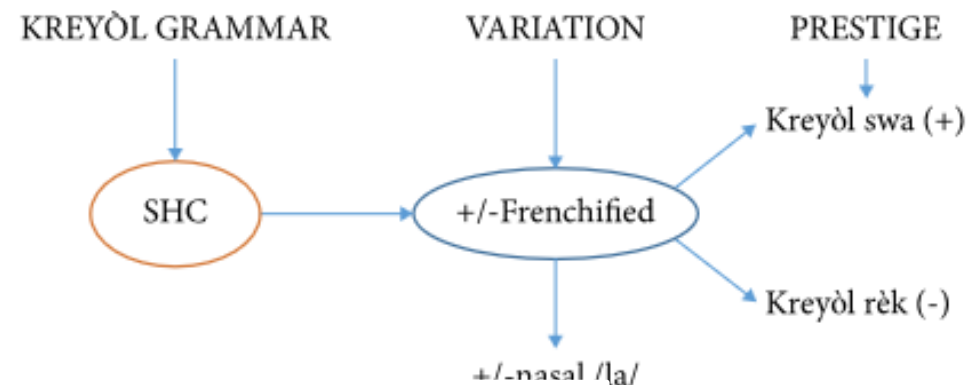
Language Barriers in Healthcare

- LEP patients face (Perez-Escamilla et al., 2010; Al Shamsi et al., 2020):
 - Misdiagnoses & treatment errors
 - Lower satisfaction & compliance
 - Higher costs & health disparities
- Current translation systems fail to address (Kreienbrinck et al., 2025):
 - Variation within Haitian Creole
 - Code-switching with French, English, Spanish
 - Lack of standardized medical terminology

Linguistic Variation in Haitian Creole

Exhibits significant linguistic (Fattier-Thomas 1984, Valdman 2015)

- **Kreyòl Swa (KS):** Spoken by bilingual Haitian speakers, more influenced by French.
 - Becoming more widespread, even among monolingual speakers (Tézil 2024).
- **Kreyòl Rèk (KR):** Spoken by monolingual Haitian Creole speakers, less influenced by French
- Haitian Creole norm based on KR
- Diaspora communities speak a wide range of varieties and possess varying degrees of knowledge of French.



Tézil 2022 p.310

Representation of KR and

1. Les douleurs musculaires avaient diminué [FR]
DET pains muscular had diminished
 2. doulè nan misk la te bese [KR HC]
pain in muscle DET.SG ANT lower
 3. doulèr muskulèr yo te diminue [KS HC]
pain muscular DET.PL ANT diminish
- « *Muscle pain had diminished* »

Tézil 2025, p.20

Lack of standardized medical terminology

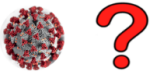
- Few Haitian Creole datasets (Mompelat 2025):
 - Small, fragmented, outdated
 - No clear distinction between varieties
 - Minimal verified medical-domain data
- Existing corpora → insufficient for healthcare NLP

Data sources for Haitian Creole

- Pre-existing available resources for HC
 - Spanish & Haitian Creole Medical Terminology: Medical Translation Pocket Booklet (Azolin et al. 2023)
 - Patient-Provider Bilingual Tools Haitian Creole and English (USSAAC 2023)

UNIVERSITY
OF MIAMI



 SUCTION ASPIRASYON	 WHAT'S MY STATUS? KISA ETA MWEN YE?	 CALL MY FAMILY RELE FANMI M	 LIGHTS ON/OFF LIMYÈ YO LIMEN/ETENN
 TROUBLE BREATHING TWOUB RESPIRASYON	 PAIN DOULÈ	 MEDICINE MEDIKAMAN	 HOT COLD CHO / FRÈT
 BATHROOM TWALÈT	 REPOSITION REPOZISYONE	 MOUTH CARE SWEN POU BOUCH	 LETTER BOARD TABLO LÈT YO
MAYBE - PETÈT		DON'T KNOW – PA KONNEN	LATER - PITA

Commands		
English	Creole	Spanish
Sit down	Chita	Siéntese
Stand up	Kanpe	Levántse
Open your eyes	Ouvri Je ou	Abra los ojos
Close your eyes	Fèmen je ou	Cierre los ojos
Turn your head	Vire tèt ou	Vire la cabeza
Open your mouth	Louvri bouch ou	Abra la boca
Stick out your tongue	Mete lang ou deyò	Saque la lengua
Lie down	Kouche	Acuéstese
Raise your hand	Leve men ou	Suba la mano
Squeeze my hand	Peze men mwen	Apriete mi mano
Raise your Legs	Leve janm ou	Suba la pierna
Take a deep breath	Pran yon gwo souf	Respire profundo

WHAT IS MY PROGNOSIS? KISA PREVIZYON MWEN YE?	WHAT ARE MY OPTIONS? KISA OPSYON MWEN YO YE?	WILL I GET BETTER? M AP FÈ MIYÒ?	AM I GOING TO DIE? ÈSKE MWEN PRAL MOURI?
WHAT WILL HAPPEN NEXT? KISA KI PRAL RIVE ANSUI?	WILL I HAVE PAIN? ÈSKE MWEN PRAL GEN DOULÈ?	I WANT TO DISCUSS MY DECISIONS MWEN VLE DISKITE SOU DESIZYON MWEN YO.	I WANT MY FAMILY TO DECIDE MWEN VLE FANMI MWEN DESIDE
WHEN WILL I COME OFF THE VENTILATOR? KILÈ MWEN AP SOT NAN RESPIRATÈ AN?	WHAT HAPPENS IF I AM TAKEN OFF THE VENTILATOR? KISA KI PRAL RIVE SI YO RETIRE M SOU RESPIRATÈ AN?	I AM NOT READY TO MAKE A DECISION MWEN PA PARE POU PRAN DESIZYON	I HAVE ANOTHER QUESTION MWEN GEN YON LÒT KESYON.
MAYBE - PETÈT		DON'T KNOW – PA KONNEN	LATER - PITA

Limitations

Datasets	Limitations
Carnegie Mellon University initiative	Approximative translations – require manual annotation and validation
Universal Dependencies treebank for Haitian Creole (Jagodźńska et al.)	Highly linguistically informative – limited quantity and genre represented
Leipzig Corpora Collection	High token quantity – unannotated data and genre-specific
Spanish & Haitian Creole Medical Terminology (Azolin et al. 2023)	Domain-specific – limited quantity and limited representation of actual language practice and use
Patient-Provider Bilingual Tools (USSAAC 2023)	

- Need for Data validation and augmentation – work in progress, in collaboration with other researchers

MT approaches

- Human translators consistently outperform Google Translate & ChatGPT in adequacy, fluency, meaning, and severity ratings (out of 5).
- ChatGPT & Google Translate show notable weaknesses in medical translation accuracy, reinforcing the need for domain-specific AI solutions.
- The end goal is to provide a tool that can assist interpreter and facilitate conversation patient/provider in absence of interpreters. (Karliner et al., 2007; Jacobs et al., 2004).

Brewster et al. 2024	Translators	GoogleTranslate	ChatGPT
Adequacy	4.5	4.0	3.9
Fluency	3.9	3.6	3.4
Meaning	4.0	3.7	3.6
Severity	4.5	4.0	3.8

Motivation for Annotation Schema

- The Problem with General-Purpose Translators
 - No guarantees of translation accuracy, especially for complex medical terms.
 - Fail to capture cultural nuances and medical literacy levels.
 - Jeopardize patient-provider communication and trust
- **Our goal:**
 - Risk-aware (NCC MERP index), variation-aware annotations
 - Foundation for safer healthcare translation



Our Approach

Using the CMU parallel corpus:

- English source → Haitian Creole MT outputs

Designed and refined annotation schema:

- 3 iterative rounds
- Native speakers + trained linguists

Designed for Haitian Creole medical translation

Annotation Schema

- Captures **5 core dimensions**:
 - **Translation Quality** – acceptable vs. improvable
 - **Register Alignment** – *kreyòl swa* vs. *kreyòl rèk* (Phase 1)
 - **Severity** – risk to patient safety
 - **Foreign Influence** – French/English intrusion (Phases 2 & 3)
 - **Error Type** – terminology, fluency, orthography

Inter-Annotator Agreement

- **3 iterative rounds** over the summer:
 - Phase 1: 300 sentences → initial schema
 - Phase 2: +150 sentences → refine definitions
 - Phase 3: +100 sentences → focus on rare, difficult cases
- Bilingual professional annotators + linguist adjudication
- Final adjudicated **gold set: 152 sentences**

Dimension	Score (k)
Severity	0.561
Quality	0.480
Fluency (error tag)	0.386
Terminology (error tag)	0.850
Foreign Influence	0.203

Challenges Encountered

- **Ambiguous terminology:** *pwatrin* vs. *lestomak*
- **Foreign Influence:** Distinguishing language-internal variation from contact-induced variation
- **Orthographic variation:** inconsistent spelling
- **High-stakes ambiguity:** dosage or procedural errors
(Solved at phase 3)

Example Annotations

- Foreign Influence = 1, quality = 0, severity = 1:
 - “Konjesyon ?” - Variant = “nen bouche”
- Foreign Influence = 1, quality = 1, severity = 0:
 - “Men ou konnen okenn lòt bagay” – Corrected sentence = “men ou pa konnen anyen ankò” :: *but you know nothing else*
- Foreign influence, severity, quality = 1
 - “Eske ou alèjik avèk sèten medikaman ?” – Corrected sentence = “Èske ou fè alèji avèk kèk medikaman” :: *Are you allergic to some medications?*

Error Types, quality = 1

- Fluency/adequacy = 1 :
 - * “Enpe la” - “enpe la a”
 - * “sansasyon sere sa a vini” - “m santi kò m vin rèd”
- Terminology = 1 :
 - # “èske ou se yon tafyatè ?” – “èske w konn bwè ?”
- Typo = 1 :
 - “èske mwen kapab repose pandan yon ti moman” - “èske mwen kapab repoze pandan yon ti moman”



Variants

- apre sa **sansasyon sere sa a vini** (quality = 1, fluency)
Corrected Sentence : apre sa, **m santi kò m vin rèd**
-> Fluency : **sansasyon sere sa a vini**=m santi kò m vin rèd
-> Variant: kò m=misk yo
- epitou èske ou bwè **alkòl** ? (quality = 1, terminology)
Corrected sentence : epitou èske ou bwè **bwason alkolize** ?
-> Terminology : **alkòl**=bwason alkolize
-> Variants : bwason alkolize=tafya; bwason alkolize=kleren
- Men ou kwè li **tèlman** grav konsa ?
Corrected sentence : men ou kwè li **reyèlman** grav konsa ?
-> Fluency : **tèlman**=**reyèlman**
-> Variant : kwè=panse

The bigger picture

- **Immediate goal:**
 - Create gold-standard evaluation set
- **Next steps:**
 - Train classifiers to flag unsafe translations
 - Reinforcement learning for MT safety
- **Long-term vision:**
 - Integrated, variation-aware MT pipeline for Haitian Creole healthcare



Conclusion

- Haitian Creole's variation and scarcity of data pose unique challenges
- Our annotation schema offers a clinically grounded foundation
- Goal: safer, more equitable healthcare communication

UNIVERSITY
OF MIAMI



Thank you 😊

- Ludovic Mompelat & Rose Flaure Accilien
- University of Miami
- lvm861@miami.edu

Find me on LinkedIn:



Ludovic Mompelat

Assistant Professor - French, Creole
and Computational Linguistics, PhD

